ENTERPRISE ENGINEERING BLUEPRINT

iMovies — Online Movie Ticket Booking Platform

Comprehensive Project Audit, Architecture Specification, Phased Implementation Roadmap & Quality Evaluation Document

Project Name	iMovies — Online Movie Ticket Booking Platform
Target Workspace	
Document Version	1.0.0 (Master Baseline Specification)
Lead Architect / PM	
Baseline Repository Audit	Physical Scan Completed: 0 Files / Greenfield Initialization
Core Technology Stack	
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iMovies — Master Project Plan, Architecture Specification & Implementation Roadmap

Project Name: iMovies — Online Movie Ticket Booking Platform
Target Repository: `d:/agent2/MovieTicket`
Document Version: 1.0.0 (Master Blueprint)
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1. Executive Summary & Role Specification

1.1 Engineering Mandate

This master plan acts as the definitive technical specification, architectural blueprint, and project roadmap for **iMovies — Online Movie Ticket Booking Platform**. Acting in the capacity of a Senior Full-Stack Software Engineer, Software Architect, Technical Project Manager, and Code Reviewer, this document provides an exhaustive, unambiguous plan for the end-to-end development, verification, security hardening, and academic/enterprise evaluation of the platform.

1.2 Core Guiding Principles

1. **Zero Assumption / Evidence-Based Auditing**: No code or component is marked as functional without verified, demonstrable evidence.

2. **Defensive Architecture**: Security, input validation, role-based authorization, and concurrency protections are designed into the core models rather than retrofitted as afterthoughts.
3. **Decoupled Modern Monolith**: A modular Express.js backend API architecture coupled with a responsive, high-performance React (Vite) frontend application.
4. **Actionability**: Every requirement is broken down into discrete, tracked work packages with unique Task IDs, explicit dependencies, measurable acceptance criteria, and effort estimates.

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2. Project Overview & Core Objectives

2.1 Mission Statement

iMovies is an enterprise-grade web application engineered to streamline the cinema discovery and ticketing lifecycle. It bridges moviegoers with theatre operators, offering frictionless browsing, real-time showtime scheduling, tier-based interactive seat selection, instant reservation management, and authentic community reviews.

2.2 Key Stakeholders & Personas

- **The Moviegoer (Customer / User)**: Discovers current and upcoming films, filters by genre and date, selects preferred seating tiers (Lower, Middle, Balcony), completes bookings, receives verifiable tickets, manages favorites, and publishes ratings/reviews.
- **The Cinema Administrator (Admin)**: Manages movie master data, configures theatres and hall capacities/pricing, schedules shows with collision detection, monitors booking volume, and moderates user feedback.
- **The System Evaluator / Auditor**: Inspects the platform for architectural integrity, database normalization, concurrency safeguards against double-booking, and compliance with industry security standards.

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3. Technology Stack Rationale & Standards

3.1 Backend Architecture

- **Runtime Environment**: Node.js (v18.x / v20.x LTS) — Asynchronous event-driven I/O optimized for concurrent HTTP requests.
- **Web Framework**: Express.js (v4.19+) — Minimalist, robust routing and middleware pipeline.
- **Database Engine**: MongoDB (v6.0+) — Document-oriented database providing schema flexibility combined with JSON-native query structures.
- **Object Data Modeling (ODM)**: Mongoose (v8.x) — Schema validation, relationship population, compound indexing, and lifecycle hooks.
- **Authentication & Cryptography**:
- `jsonwebtoken` (v9.x): Stateless JWT generation with HS256 cryptographic signing and explicit expiration.
- `bcryptjs` (v2.4.3): Adaptive one-way salted password hashing (work factor / salt rounds >= 10).
- **Validation & Security**:
- `express-validator` : Declarative request payload validation and sanitization.
- `cors` : Explicit origin-whitelisted Cross-Origin Resource Sharing.
- `helmet` : HTTP header hardening.
- `express-rate-limit` : Brute-force and DDoS mitigation on sensitive authentication endpoints.

3.2 Frontend Architecture

- **Framework & Build Tool**: React.js (v18.3+) with Vite (v5.x) — Instant Hot Module Replacement (HMR) and optimized Rollup production bundling.
- **Client Routing**: React Router DOM (v6.x) — Declarative nested routing, layout wrappers, and protected route guards.

- **HTTP Client**: Axios (v1.7+) — Unified instance with request/response interceptors for automatic JWT injection and centralized 401 handling.
- **UI & Styling**: Pure Vanilla Modern CSS — Custom CSS design tokens (variables), glassmorphic dark-theme aesthetics, responsive grid layouts, and zero heavy external component library lock-in.
- **Iconography**: Lucide React — Lightweight tree-shakable SVG icons.

3.3 Port Allocations & Network Topology

- **Backend API Service**: `http://localhost:5000` (Reverse-proxied in production or configured via CORS)
- **Frontend Development Server**: `http://localhost:5173`
- **Database Instance**: `mongodb://localhost:27017/imovies_db` (or MongoDB Atlas URI)

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4. Rigorous Codebase Audit & Baseline State Analysis

4.1 Repository Physical Inspection

An inspection of the workspace path `d:\agent2\MovieTicket` was executed.

```
[AUDIT LOG: PHYSICAL REPOSITORY SCAN]
Target Path: d:\agent2\MovieTicket
Items Discovered on Disk: 0 files, 0 folders (prior to /docs initialization)
Neighboring Directory Scan: d:\agent2\* searched for references to iMovies/MovieTicket
Findings: No active source code, models, routes, or configuration files exist in the workspace.
```

4.2 Audit Classification Matrix

To satisfy the audit requirements with zero fabrication, every component referenced in the initial project brief has been audited against real disk evidence:

Expected Component	Referenced File Path	Real Status on Disk	Architectural & Security Findings	Action Required
Package Manifest	`backend/package.json`	**Not implemented**	No dependency manifests or scripts present.	Initialize clean Node.js package with locked dependencies.
Server Entry Point	`backend/server.js`	**Not implemented**	No Express instance, DB connector, or middleware pipeline exists.	Create modular `server.js` with graceful shutdown and error middleware.
Authentication Middleware	`backend/middlewares/fetchUser.js`	**Not implemented**	Referenced in notes, but missing. Initial design conflated auth and role checks.	Implement separate `authenticateUser` and `authorizeRole('admin')` middlewares.
User Model	`backend/models/userModel.js`	**Not implemented**	Entity not created.	Define Mongoose schema with email uniqueness, lowercase trimming, and password hashing hooks.
Admin Model	`backend/models/adminModel.js`	**Not implemented**	Architectural anti-pattern: Separate admin model causes duplicate auth logic and schema redundancy.	Unify into `userModel.js` with `role: { type: String, enum: ['user', 'admin'], default: 'user' }`.
Movie Model	`backend/models/movieModel.js`	**Not implemented**	Entity not created.	Define Movie schema with certification enums, runtime, media URLs, and text indexes.
Theatre Model	`backend/models/theatreModel.js`	**Not implemented**	Entity not created. ER diagram notes string admin name.	Define Theatre schema using `ObjectId` reference to Admin, plus tier seat capacities.
Show Model	`backend/models/showModel.js`	**Not implemented**	Entity not created.	Define Show schema with compound index (`theatre + date + time`), movie reference, and booked seats array.
Booking Model	`backend/models/bookingModel.js`	**Not implemented**	Entity not created.	Define Booking schema with atomic status, ticket details, UUID reference, and user/show references.
Favorite Model	`backend/models/favoriteModel.js`	**Not implemented**	Entity not created.	Define Favorite schema with unique compound index on `(user, movie)` to prevent duplicates.
Review Model	`backend/models/reviewModel.js`	**Not implemented**	Entity not created.	Define Review schema with unique compound index `(user, movie)` and rating bounds [1, 5].
Route Modules	`backend/routes/*.js` (8 files)	**Not implemented**	Zero route files present.	Implement clean REST controllers using Express Router with validator middlewares.
Frontend Application	`frontend/*`	**Not implemented**	No client application found.	Scaffold clean React + Vite application with responsive CSS design tokens.
Automated Tests	`backend/tests/*`	**Not implemented**	No unit or integration tests exist.	Configure Jest, Supertest, and MongoMemoryServer for automated CI/CD validation.

5. Domain Model, ER Diagram & Schema Architecture

5.1 Analysis of Original ER Diagram vs Recommended Mongoose Modeling

The initial domain model features seven primary entities: User, Admin, Movie, Theatre, Show, Booking, and Favorite. A rigorous architectural review reveals critical areas requiring modernization and structural enhancement:

1. **User vs Admin Entity Segregation**:
2. *Original Diagram*: Separate `User` and `Admin` entities with identical attributes (`Username` , `Email` , `Password`).
3. *Architectural Recommendation*: **Unify into a single `User` model with a `role` discriminator (`'user' | 'admin'`)**. Maintaining separate tables causes duplicate authentication logic, split JWT validation paths, and redundant token middleware.
4. **Entity Reference Types (Strings vs ObjectIds)**:
5. *Original Diagram*: Theatre references Admin via string name; Show references Theatre via string name; Booking references User via email string.
6. *Architectural Recommendation*: **Replace all loose string references with canonical MongoDB `Schema.Types.ObjectId` references with `ref` declarations**. String references break referential integrity if a user changes their email or a theatre is renamed.
7. **Seat Availability & Capacity Modeling**:
8. *Original Diagram*: Theatre defines `lower-seat count` , `middle-seat count` , and `balcony-seat count` with seat prices.
9. *Architectural Recommendation*:
    - **Theatre Model**: Defines hall layout, tier capacities (`lowerCount` , `middleCount` , `balconyCount`), and base tier prices.
    - **Show Model**: Overrides base prices if required and maintains an atomic array of `bookedSeats: [String]` (e.g. `['L-1', 'L-2', 'M-12']`).
    - **Booking Model**: Records the exact seats booked, tier breakdown, total amount paid, and booking reference UUID.
10. **Uniqueness & Referential Constraints**:
11. Compound unique index on `Favorite` : `{ user: 1, movie: 1 }` (A user can only favorite a movie once).
12. Compound unique index on `Review` : `{ user: 1, movie: 1 }` (A user can only review a movie once; updates allowed via PUT).
13. Compound scheduling index on `Show` : `{ theatre: 1, showDate: 1, showTime: 1 }` to prevent overlapping hall reservations.

5.2 Mongoose Schema Definitions (Architectural Blueprint)

```
// ========================================
// 1. USER SCHEMA (Unified with RBAC)
// ========================================
const userSchema = new mongoose.Schema({
  username: { type: String, required: true, trim: true, minlength: 3, maxlength: 50 },
  email: { type: String, required: true, unique: true, lowercase: true, trim: true, match: [/^\S+@\S+\.\S+$/, 'Invalid email'] },
  password: { type: String, required: true, minlength: 8 },
  role: { type: String, enum: ['user', 'admin'], default: 'user', index: true }
}, { timestamps: true });

// ========================================
// 2. MOVIE SCHEMA
// ========================================
const movieSchema = new mongoose.Schema({
  title: { type: String, required: true, trim: true, index: 'text' },
  description: { type: String, required: true },
  genre: [{ type: String, required: true, trim: true }],
  certification: { type: String, enum: ['U', 'UA', 'A', 'R'], required: true },
  runtimeMinutes: { type: Number, required: true, min: 1 },
  releaseDate: { type: Date, required: true },
  posterUrl: { type: String, required: true },
  backdropUrl: { type: String },
  trailerUrl: { type: String },
  averageRating: { type: Number, default: 0, min: 0, max: 5 },
  reviewCount: { type: Number, default: 0 }
}, { timestamps: true });

// ========================================
// 3. THEATRE SCHEMA
// ========================================
const theatreSchema = new mongoose.Schema({
  name: { type: String, required: true, trim: true },
  location: { type: String, required: true, trim: true },
  city: { type: String, required: true, trim: true, index: true },
  admin: { type: mongoose.Schema.Types.ObjectId, ref: 'User', required: true },
  seatingCapacity: {
    lower: { count: { type: Number, required: true, min: 1 }, defaultPrice: { type: Number, required: true, min: 0 } },
    middle: { count: { type: Number, required: true, min: 1 }, defaultPrice: { type: Number, required: true, min: 0 } },
    balcony: { count: { type: Number, required: true, min: 1 }, defaultPrice: { type: Number, required: true, min: 0 } }
  }
}, { timestamps: true });

// ========================================
// 4. SHOW SCHEMA
// ========================================
const showSchema = new mongoose.Schema({
  movie: { type: mongoose.Schema.Types.ObjectId, ref: 'Movie', required: true, index: true },
  theatre: { type: mongoose.Schema.Types.ObjectId, ref: 'Theatre', required: true, index: true },
  showDate: { type: String, required: true, index: true }, // Format: YYYY-MM-DD
  showTime: { type: String, required: true },               // Format: HH:mm
  pricing: {
    lower: { type: Number, required: true, min: 0 },
    middle: { type: Number, required: true, min: 0 },
    balcony: { type: Number, required: true, min: 0 }
  },
  bookedSeats: [{ type: String }], // Array of seat IDs, e.g., ['L-1', 'M-5', 'B-10']
  totalCapacity: { type: Number, required: true }
}, { timestamps: true });

showSchema.index({ theatre: 1, showDate: 1, showTime: 1 }, { unique: true });

// ========================================
// 5. BOOKING SCHEMA
// ========================================
const bookingSchema = new mongoose.Schema({
  bookingReference: { type: String, required: true, unique: true, index: true }, // UUID or alphanumeric
  user: { type: mongoose.Schema.Types.ObjectId, ref: 'User', required: true, index: true },
  show: { type: mongoose.Schema.Types.ObjectId, ref: 'Show', required: true, index: true },
```

```
  seats: [{
    seatNumber: { type: String, required: true },
    tier: { type: String, enum: ['lower', 'middle', 'balcony'], required: true },
    price: { type: Number, required: true, min: 0 }
  }],
  totalAmount: { type: Number, required: true, min: 0 },
  status: { type: String, enum: ['CONFIRMED', 'CANCELLED'], default: 'CONFIRMED', index: true },
  bookingDate: { type: Date, default: Date.now }
}, { timestamps: true });

// ========================================
// 6. FAVORITE SCHEMA
// ========================================
const favoriteSchema = new mongoose.Schema({
  user: { type: mongoose.Schema.Types.ObjectId, ref: 'User', required: true },
  movie: { type: mongoose.Schema.Types.ObjectId, ref: 'Movie', required: true }
}, { timestamps: true });

favoriteSchema.index({ user: 1, movie: 1 }, { unique: true });

// ========================================
// 7. REVIEW SCHEMA
// ========================================
const reviewSchema = new mongoose.Schema({
  user: { type: mongoose.Schema.Types.ObjectId, ref: 'User', required: true },
  movie: { type: mongoose.Schema.Types.ObjectId, ref: 'Movie', required: true, index: true },
  rating: { type: Number, required: true, min: 1, max: 5 },
  comment: { type: String, required: true, trim: true, maxlength: 1000 }
}, { timestamps: true });

reviewSchema.index({ user: 1, movie: 1 }, { unique: true });
```

---

6. Functional Requirements & Epic Breakdown

Epic 1: Authentication, Authorization & Identity Management

- **User Self-Registration**: Public endpoint permitting new customers to create an account with username, email, and strong password. Password hashed with bcrypt before persistence.
- **Unified Login Service**: Generates signed JWT containing `userId` and `role`. Strips password hash from all JSON output.
- **Middleware Layer**:
- `fetchUser`: Verifies Authorization header `Bearer <token>`, validates expiration, attaches `req.user`.
- `authorizeRole(roles)`: Enforces role checks (`'admin'`), rejecting non-authorized users with HTTP 403 Forbidden.
- **Profile Management**: Authenticated users can retrieve and update their personal profile.

Epic 2: Movies & Cinema Discovery

- **Public Movie Catalog**: Displays currently screening and upcoming movies with high-resolution poster media, certification badges, runtime, and genre tags.
- **Search & Filtering Engine**: Text search on title/description and multi-genre filtering with pagination support (`page` , `limit`).
- **Movie Details View**: In-depth view featuring movie synopsis, runtime, certification, trailer link, community rating, and active show schedules.
- **Admin Movie Management**: Protected administrative endpoints to create, update, or soft-delete films.

Epic 3: Theatre & Show Schedule Management

- **Theatre Infrastructure Configuration**: Admins register theatre locations, hall details, and tier seating capacities (Lower, Middle, Balcony).

- **Show Scheduling Engine**: Connects a Movie, Theatre, Date, and Time. Implements conflict validation to prevent double-booking a theatre hall during overlapping timeslots.
- **Showtime Discovery**: Query shows by `movieId` and `date`, grouped cleanly by theatre with remaining seat tallies.

Epic 4: Interactive Seat Availability & Atomic Booking

- **Visual Tiered Seat Map**: Visualizes the hall layout partitioned into Lower, Middle, and Balcony tiers, dynamically displaying booked versus available seats.
- **Atomic Reservation & Double-Booking Prevention**: Executes atomic reservation using MongoDB conditional queries (`$nin` on `bookedSeats` array) to guarantee zero double-booking even under extreme concurrent requests.
- **Ticket Generation**: Issues an immutable booking confirmation with reference code, seat identifiers, showtime, and total price.
- **Customer Booking History & Cancellation**: View active and historical bookings with ticket print view and pre-show cancellation options.

Epic 5: Favorites & Movie Reviews

- **User Watchlist / Favorites**: Allows logged-in users to toggle films into their personal favorites list with instant UI reflection.
- **Community Ratings & Reviews**: Enables customers to post verified star ratings (1 to 5) and textual reviews. Enforces single review per user per movie with recalculation of movie `averageRating`.

Epic 6: Administration, Reporting & Platform Operations

- **Administrator Dashboard**: Aggregated operational metrics: total registered users, active shows, tickets sold, and revenue breakdown.
- **Schedule & Ticket Oversight**: Admins can audit all platform bookings, filter by date/theatre, and manage scheduling exceptions.

---

7. Complete REST API Inventory & Contract Specifications

7.1 Unified API Response Standard

All backend endpoints strictly adhere to a standardized JSON response envelope:

```
// Success Response (HTTP 200 / 201)
{
  "success": true,
  "message": "Operation completed successfully.",
  "data": { ... }
}

// Error Response (HTTP 400 / 401 / 403 / 404 / 409 / 500)
{
  "success": false,
  "message": "Detailed human-readable error description.",
  "error": {
    "code": "ERROR_CODE_IDENTIFIER",
    "details": [ ... ]
  }
}
```

7.2 Detailed API Endpoint Inventory

Endpoint Route	HTTP	Purpose	Auth	Role	Request Body / Params	Expected Responses	Status
`/api/auth/register`	POST	Register new customer	None	Public	`{ username, email, password }`	201 (Created), 400 (Validation), 409 (Email Exists)	Proposed
`/api/auth/login`	POST	Authenticate user	None	Public	`{ email, password }`	200 (JWT + user object), 400 (Bad input), 401 (Invalid creds)	Proposed
`/api/auth/profile`	GET	Fetch active profile	Yes	User/Admin	None	200 (User details sans password), 401 (Unauthorized)	Proposed
`/api/auth/profile`	PUT	Update user profile	Yes	User/Admin	`{ username }`	200 (Updated user), 400 (Validation), 401 (Unauthorized)	Proposed
`/api/movies`	GET	List & search movies	None	Public	Query: `?genre=&search=&page=&limit=`	200 (Paginated list of movies, total, pages)	Proposed
`/api/movies/:id`	GET	Retrieve movie by ID	None	Public	Param: `id` (ObjectId)	200 (Movie details), 400 (Invalid ID), 404 (Not found)	Proposed
`/api/movies`	POST	Create new movie	Yes	Admin	`{ title, description, genre, certification, runtimeMinutes, releaseDate, posterUrl }`	201 (Created movie), 400 (Validation), 403 (Forbidden)	Proposed
`/api/movies/:id`	PUT	Update movie details	Yes	Admin	Param: `id`, Body: Movie fields	200 (Updated movie), 400 (Validation), 404 (Not found)	Proposed
`/api/movies/:id`	DELETE	Delete/Archive movie	Yes	Admin	Param: `id`	200 (Success message), 404 (Not found), 409 (Active shows exist)	Proposed
`/api/theatres`	GET	List all theatres	None	Public	Query: `?city=`	200 (Array of theatres with seating configs)	Proposed

Endpoint Route	HTTP	Purpose	Auth	Role	Request Body / Params	Expected Responses	Status
`/api/theatres/:id`	GET	Theatre details	None	Public	Param: `id`	200 (Theatre object), 404 (Not found)	Proposed
`/api/theatres`	POST	Create new theatre	Yes	Admin	`{ name, location, city, seatingCapacity }`	201 (Created theatre), 400 (Validation), 403 (Forbidden)	Proposed
`/api/theatres/:id`	PUT	Update theatre	Yes	Admin	Param: `id`, Body: Theatre fields	200 (Updated theatre), 400 (Validation), 404 (Not found)	Proposed
`/api/shows`	GET	Query active shows	None	Public	Query: `?movieId=&theatreId=&date=`	200 (List of matching shows populated with theatre info)	Proposed
`/api/shows/:id`	GET	Show details & pricing	None	Public	Param: `id`	200 (Show details, pricing, booked seat IDs)	Proposed
`/api/shows/:id/seats`	GET	Live seat map status	None	Public	Param: `id`	200 (Tier layouts, booked seat list, remaining counts)	Proposed
`/api/shows`	POST	Schedule new show	Yes	Admin	`{ movie, theatre, showDate, showTime, pricing }`	201 (Created show), 400 (Validation), 409 (Hall schedule conflict)	Proposed
`/api/shows/:id`	DELETE	Cancel/Remove show	Yes	Admin	Param: `id`	200 (Success), 409 (Show has existing confirmed bookings)	Proposed
`/api/bookings`	POST	Book seats atomically	Yes	User	`{ showId, selectedSeats: ['L-1', 'L-2'] }`	201 (Confirmed booking & ticket), 400 (Validation), 409 (Seat already booked)	Proposed
`/api/bookings/my-bookings`	GET	User booking history	Yes	User	None	200 (Array of user bookings with show & movie info)	Proposed

Endpoint Route	HTTP	Purpose	Auth	Role	Request Body / Params	Expected Responses	Status
`/api/bookings/:id`	GET	Booking ticket details	Yes	User/Admin	Param: `id`	200 (Detailed ticket view), 403 (Access denied), 404 (Not found)	Proposed
`/api/bookings/:id/cancel`	PUT	Cancel user booking	Yes	User	Param: `id`	200 (Cancelled & seats released), 400 (Past show cutoff)	Proposed
`/api/favorites`	GET	Fetch user favorites	Yes	User	None	200 (Array of favorited movies)	Proposed
`/api/favorites/:movieId`	POST	Add movie to favorites	Yes	User	Param: `movieId`	201 (Added to favorites), 409 (Already favorited)	Proposed
`/api/favorites/:movieId`	DELETE	Remove from favorites	Yes	User	Param: `movieId`	200 (Removed), 404 (Not found in favorites)	Proposed
`/api/movies/:id/reviews`	GET	Get reviews for movie	None	Public	Param: `id`	200 (Array of reviews with author username)	Proposed
`/api/reviews`	POST	Submit movie review	Yes	User	`{ movieId, rating, comment }`	201 (Review created & movie average updated), 409 (Duplicate review)	Proposed
`/api/reviews/:id`	PUT	Update user review	Yes	User	Param: `id` , Body: `{ rating, comment }`	200 (Updated review & recomputed average), 403 (Not review owner)	Proposed
`/api/reviews/:id`	DELETE	Delete review	Yes	User/Admin	Param: `id`	200 (Review removed), 403 (Not owner or admin)	Proposed
`/api/admin/stats`	GET	Platform analytics	Yes	Admin	None	200 (Total revenue, user count, tickets sold, active shows)	Proposed
`/api/admin/bookings`	GET	Audit all bookings	Yes	Admin	Query: `?status=&theatreId=&date=`	200 (Paginated list of all system bookings)	Proposed

8. Frontend Architecture & Page-by-Page Implementation Plan

8.1 Styling Strategy & Architectural Rationale

In accordance with modern web engineering best practices: - **Primary Styling**: Pure Vanilla Modern CSS utilizing custom CSS design variables (`:root`) for color palettes, spacing tokens, elevations, and responsive typography. This provides total layout control, zero third-party component lock-in, and instant load times. - **Theme Palette**: Deep cinematic dark mode (Background: `#0B0F19` , Surface: `#161F30` , Card: `#1F293D` , Accent Amber: `#F59E0B` , Vibrant Cyan: `#06B6D4` , Crimson Alert: `#EF4444`). - **Responsive Breakpoints**: Mobile (`< 640px`), Tablet (`640px - 1024px`), Desktop (`> 1024px`).

8.2 Client Route Structure

```
/                     -> HomePage (Hero banner, now showing, upcoming)
/movies               -> MovieCatalogPage (Search, genre filters, movie grid)
/movies/:id           -> MovieDetailPage (Metadata, synopsis, trailer, date picker, showtimes)
/book/:showId         -> SeatBookingPage (Interactive seat layout, tier pricing, live checkout)
/booking/success/:id  -> BookingConfirmationPage (Ticket slip, QR placeholder, print view)
/login                -> LoginPage (Customer & Admin authentication)
/register             -> RegisterPage (Account creation)
/profile              -> UserProfilePage (Account settings, preferences)
/my-bookings          -> MyBookingsPage (Past & active tickets, cancellation)
/favorites            -> FavoritesPage (User's saved movies)
/admin/dashboard      -> AdminDashboardPage (Analytics, revenue counters)
/admin/movies         -> AdminMoviesPage (Movie master catalog management)
/admin/theatres       -> AdminTheatresPage (Theatre capacity and tier setup)
/admin/shows          -> AdminShowsPage (Showtime scheduler with collision detection)
/admin/bookings       -> AdminBookingsPage (Platform booking audit log)
```

8.3 Detailed Page Specifications

Page 1: HomePage (`/`)

- **Purpose**: Welcoming landing portal showcasing trending films, quick search, and active cinema screenings.
- **Main Components**: HeroCarousel, QuickSearchWidget, NowShowingSection, UpcomingReleasesSection, Footer.
- **Backend Endpoints**: `GET /api/movies?limit=8` .
- **States**: Skeleton cards on load, empty state banner if no screenings exist, error toast on network failure.
- **Acceptance Criteria**: Displays movie poster cards with ratings, clicking a card routes to `/movies/:id` .

Page 2: MovieCatalogPage (`/movies`)

- **Purpose**: Full discovery page allowing multi-criteria filtering and text search.
- **Main Components**: FilterSidebar (genres, certifications), SearchBar (debounced input), MovieGrid, PaginationControls.
- **Backend Endpoints**: `GET /api/movies?search=...&genre=...&page=...` .
- **States**: Live loading spinner, zero search results fallback screen with reset button.

Page 3: MovieDetailPage (`/movies/:id`)

- **Purpose**: Comprehensive movie profile and cinema showtime selector.
- **Main Components**: BackdropHero, MovieMetadataCard, SynopsisBlock, DateSelectorBar (next 7 days), TheatreShowtimeAccordion, ReviewsList, ReviewSubmissionModal.
- **Backend Endpoints**: `GET /api/movies/:id` , `GET /api/shows?movieId=:id&date=YYYY-MM-DD` , `GET /api/movies/:id/reviews` , `POST /api/favorites/:id` .
- **Acceptance Criteria**: Selecting a date dynamically updates the showtime buttons grouped by theatre. Clicking a showtime routes to `/book/:showId` .

Page 4: Interactive SeatBookingPage (`/book/:showId`)

- **Purpose**: High-fidelity interactive cinema hall seat selector with tier segregation.
- **Main Components**: ScreenCurveVisual, TierHeader (Lower, Middle, Balcony with pricing), SeatGrid, SeatLegend (Available, Selected, Booked), OrderSummaryPanel, ConfirmBookingButton.
- **Backend Endpoints**: `GET /api/shows/:id/seats`, `POST /api/bookings`.
- **User Interactions**:
- Clicking an available seat toggles selection state.
- Booked seats are permanently disabled with visual lock styling.
- Dynamic checkout panel calculates subtotal across multiple tiers.
- Submitting dispatches atomic booking request. On 409 Conflict (seat taken concurrently), alerts user and refreshes seat map.
- **Acceptance Criteria**: Strictly enforces that already booked seats cannot be clicked. Successful booking routes to `/booking/success/:id`.

Page 5: BookingConfirmationPage (`/booking/success/:id`)

- **Purpose**: Digital ticket receipt with print functionality.
- **Main Components**: TicketCard (Movie poster, theatre name, hall, show date/time, seat codes, total price, booking reference UUID), PrintTicketButton, BackToHomeButton.
- **Backend Endpoints**: `GET /api/bookings/:id`.
- **Acceptance Criteria**: Formatted for browser print (`@media print` stylesheet) and accessible in customer's booking history.

---

9. Security, Concurrency & Data Integrity Strategy

9.1 Concurrency & Double-Booking Prevention Architecture

In online ticket booking systems, race conditions where two simultaneous requests attempt to reserve the same seat (`L-15`) represent the highest risk failure mode.

The Problem with Naive Implementation:

```
// NAIVE (BUGGY) APPROACH - LEADS TO DOUBLE BOOKING:
const show = await Show.findById(showId);
if (show.bookedSeats.includes('L-15')) {
  return res.status(400).send("Seat already booked");
}
// Concurrency window: Another request executes here before save!
show.bookedSeats.push('L-15');
await show.save();
```

Approved Production Solution: Database-Level Atomic Conditional Updates

We utilize MongoDB's atomic single-document conditional update operators:

```
// ATOMIC & RACE-CONDITION-FREE BOOKING LOGIC:
const updatedShow = await Show.findOneAndUpdate(
  {
    _id: showId,
    // Condition: ALL requested seats must NOT be present in bookedSeats
    bookedSeats: { $nin: selectedSeats }
  },
  {
    // Atomically push all requested seats into the booked array
    $push: { bookedSeats: { $each: selectedSeats } }
  },
  { new: true }
);

if (!updatedShow) {
  // If no document matched, at least one seat was claimed concurrently
  return res.status(409).json({
    success: false,
    message: "One or more selected seats have already been booked by another customer. Please choose different seats.",
    error: { code: "SEATS_CONCURRENTLY_RESERVED" }
  });
}

// Seat reservation secured atomically; create confirmed booking record
const booking = await Booking.create({
  bookingReference: `IMV-${Date.now()}-${Math.floor(1000 + Math.random() * 9000)}`,
  user: req.user.id,
  show: showId,
  seats: seatDetailsArray,
  totalAmount: calculatedTotal,
  status: 'CONFIRMED'
});
```

9.2 Cryptographic Controls & Authentication Security

- **Bcrypt Salt Generation**: `bcrypt.hash(password, 10)`. Passwords are never stored in plaintext or logged.
- **Password Exclusion in Serialization**: The User schema utilizes Mongoose's `toJSON` transform to guarantee `password` is never included in outgoing responses: `javascript userSchema.set('toJSON', { transform: (doc, ret) => { delete ret.password; return ret; } });`
- **JWT Lifecycles & Security**: Signed using a minimum 256-bit secret key with an expiration window of 24 hours (`expiresIn: '24h'`).
- **NoSQL Injection Defense**: Strict type validation through `express-validator` prevents query object injection attacks (e.g. `{ "email": { "$gt": "" } }`).
- **CORS Protection**: CORS configured to reject arbitrary external origins, explicitly binding to authorized client domains.
- **Rate Limiting**: Configured using `express-rate-limit` allowing maximum 10 login attempts per 15-minute window per IP to eliminate brute-force vulnerability.

---

10. Comprehensive Testing Strategy & Test Matrices

10.1 Automated Testing Architecture

Testing is structured across three concentric tiers using **Jest**, **Supertest**, and **mongodb-memory-server**: 1. **Unit Tests**: Pure business logic (seat pricing calculation, JWT helper functions, password hashing utilities). 2. **Integration Tests**: REST API endpoints, middleware authorization pipelines, and Mongoose query transactions. 3. **Concurrency Stress Tests**: Simulated high-throughput race condition scripts.

10.2 Concurrency Stress Test Specification (`CONC-TEST-01`)

- **Objective**: Mathematically prove zero double-booking under concurrent load.
- **Methodology**:
- Initialize test Show with 100 available seats.

- Launch 10 asynchronous concurrent HTTP `POST /api/bookings` requests targeting identical seat `L-5` simultaneously using `Promise.all()`.
- **Expected Outcome**: Exactly 1 request receives HTTP 201 Created; exactly 9 requests receive HTTP 409 Conflict.
- Inspect database: `Show.bookedSeats` contains exactly one entry for `L-5`.

10.3 Negative & Security Test Suite Matrix

Test ID	Target Component	Test Scenario	Input Payload	Expected HTTP Status	Expected Assertion
NEG-01	`POST /api/auth/register`	Duplicate Email Registration	Existing user email	`409 Conflict`	Message indicates email already registered
NEG-02	`POST /api/auth/register`	Weak Password	Password `< 8` chars	`400 Bad Request`	Validation error array identifies password rule
NEG-03	`POST /api/auth/login`	Invalid Password	Correct email, wrong password	`401 Unauthorized`	Generic credential rejection message
NEG-04	`GET /api/admin/stats`	Regular User Accessing Admin	Valid User JWT token	`403 Forbidden`	Access denied for non-admin role
NEG-05	`GET /api/movies/:id`	Invalid MongoDB ObjectId	`GET /api/movies/invalid-id-123`	`400 Bad Request`	Handled gracefully without CastError crash
NEG-06	`POST /api/shows`	Overlapping Show Schedule	Same theatre, overlapping time	`409 Conflict`	Schedule conflict detected
NEG-07	`POST /api/bookings`	Booking Without Token	Missing Authorization header	`401 Unauthorized`	Access token required
NEG-08	`PUT /api/bookings/:id/cancel`	Cancelling Past Show	Booking ID for show from yesterday	`400 Bad Request`	Cancellation window expired

---

11. Phased Implementation Roadmap & Dependency Matrix

```
graph TD
    P0[Phase 0: Audit & Specification] --> P1[Phase 1: Environment & Base Setup]
    P1 --> P2[Phase 2: Database Models & Schemas]
    P2 --> P3[Phase 3: Auth & Identity Services]
    P3 --> P4[Phase 4: Movie Catalog APIs]
    P3 --> P5[Phase 5: Theatres & Show Scheduler]
    P4 --> P6[Phase 6: Seat Booking & Concurrency]
    P5 --> P6
    P4 --> P7[Phase 7: Favorites & Reviews]
    P6 --> P8[Phase 8: Frontend Application]
    P7 --> P8
    P8 --> P9[Phase 9: Testing & Security Audit]
    P9 --> P10[Phase 10: Deployment & Handover]
```

11.1 Detailed Phase Breakdown

Phase 0: Repository Audit & Requirements Baseline (Completed)

- **Objective**: Conduct exhaustive audit of workspace, identify gaps, and publish architectural blueprints.
- **Deliverables**: `docs/iMovies_Project_Plan.md`, `docs/iMovies_Evaluation_Checklist.md`, `docs/iMovies_Task_Tracker.csv`, `docs/iMovies_Project_Plan.pdf`.

Phase 1: Environment & Server Foundation (Effort: 4-6 Hours)

- **Tasks**: `ENV-01`, `ENV-02`, `ENV-03`.
- **Deliverables**: Initialized `package.json`, Express application entry (`server.js`), MongoDB connection module with event listeners, `.env.example`, security headers.
- **Dependencies**: None.

Phase 2: Schema Architecture & Data Integrity (Effort: 6-8 Hours)

- **Tasks**: `AUTH-01`, `MOV-01`, `THTR-01`, `SHOW-01`, `BOOK-01`, `FAV-01`, `REV-01`.
- **Deliverables**: All 7 Mongoose models implemented with field constraints, compound indexes, and relationship references.
- **Dependencies**: Phase 1.

Phase 3: Authentication & Role Authorization (Effort: 6-8 Hours)

- **Tasks**: `AUTH-02`, `AUTH-03`, `AUTH-04`, `AUTH-05`, `AUTH-06`, `AUTH-07`, `AUTH-08`.
- **Deliverables**: User/Admin registration, login, JWT issuance, `fetchUser` and `authorizeRole` middlewares.
- **Dependencies**: Phase 2.

Phase 4: Movies Discovery & Master Catalog (Effort: 5-7 Hours)

- **Tasks**: `MOV-02`, `MOV-03`, `MOV-04`, `MOV-05`.
- **Deliverables**: Public movie catalog search, genre filtering, pagination, and admin CRUD endpoints.
- **Dependencies**: Phase 3.

Phase 5: Theatre & Show Scheduling Engine (Effort: 6-8 Hours)

- **Tasks**: `THTR-02`, `SHOW-02`, `SHOW-03`.
- **Deliverables**: Theatre management APIs, collision-free show scheduling, showtimes query by date.
- **Dependencies**: Phase 4.

Phase 6: Seat Availability & Atomic Booking Engine (Effort: 8-10 Hours)

- **Tasks**: `BOOK-02`, `BOOK-03`, `BOOK-04`, `BOOK-05`.
- **Deliverables**: Live seat map API, atomic conditional booking engine, user booking history, ticket cancellation.
- **Dependencies**: Phase 5.

Phase 7: Community Reviews & Favorites (Effort: 4-6 Hours)

- **Tasks**: `FAV-01`, `REV-02`.
- **Deliverables**: Movie favorites toggle, review submission and rating recalculation.
- **Dependencies**: Phase 4.

Phase 8: Frontend Client Development (Effort: 16-20 Hours)

- **Tasks**: `UI-01` through `UI-09`.
- **Deliverables**: Full React + Vite application with responsive CSS design tokens, AuthContext, interactive seat selector, booking flow, and admin portal.
- **Dependencies**: Phases 3, 4, 5, 6, 7.

Phase 9: Automated Testing & Security Hardening (Effort: 8-10 Hours)

- **Tasks**: `QA-01`, `QA-02`, `QA-03`.

- **Deliverables**: Jest test suites, automated double-booking concurrency test, security audit report.
- **Dependencies**: Phase 8.

Phase 10: Deployment, Documentation & Evaluation (Effort: 4-6 Hours)

- **Tasks**: `DEP-01`.
- **Deliverables**: Production build, Dockerfile/compose configuration, final evaluation audit walkthrough.
- **Dependencies**: Phase 9.

---

12. Prioritization, Scope Control & MVP Definition

12.1 Priority Classifications

- **P0 (Critical Path / MVP Core)**: Must be 100% operational for the application to function.
- User Registration & Login with JWT and Bcrypt
- Movie Catalog, Details & Search
- Theatre & Show Creation with Conflict Prevention
- Real-Time Seat Map Display
- Atomic Seat Booking with Race-Condition Prevention
- User Ticket View & Confirmation
- **P1 (Standard Release Baseline)**: Required for a polished full release.
- Movie Favorites Management
- User Reviews & Star Ratings
- Booking Cancellation & Seat Release
- Admin Dashboard & Analytics Counters
- Responsive Mobile Navigation
- **P2 (Useful Enhancements)**: Post-launch iterative improvements.
- Razorpay / Stripe Payment Gateway Integration
- Asynchronous Email Ticket Confirmation (Nodemailer)
- Real-Time Live Seat Locks via WebSockets (Socket.IO)
- **P3 (Future Extensions)**: Enterprise scaling.
- Microservices decoupling, Redis caching layer, multi-cinema chain SaaS tenancy.

---

13. Project Evaluation Rubric & Quality Assurance Standards

13.1 Scoring Distribution (100 Points Total)

- **Functional Correctness & Completeness**: 20 Points
- **Backend Architecture & Code Quality**: 15 Points
- **Database Design, Relationships & Data Integrity**: 15 Points
- **Authentication, Authorization & Security**: 15 Points
- **Booking Concurrency & Seat Integrity**: 10 Points
- **Frontend Usability, Aesthetics & Responsiveness**: 10 Points
- **Automated Testing & Error Handling**: 10 Points
- **Documentation, Deployment & Repository Hygiene**: 5 Points

(Refer to `docs/iMovies_Evaluation_Checklist.md` for the line-item grading sheet and measurable evidence criteria).

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14. Comprehensive Risk Register

Risk ID	Risk Description	Prob.	Impact	Severity	Affected Component	Mitigation Strategy	Verification Method
RSK-01	Concurrent double booking of identical seats during high-demand release	High	Critical	**High**	Booking Engine / Show Model	Implement database-level atomic conditional update (`$nin` operator)	Automated concurrency test script with 10 simultaneous requests
RSK-02	Conflicting show schedules assigned to same theatre hall	Med	High	**High**	Show Scheduling API	Compound unique index `(theatre, date, time)` + duration collision check	Attempt creating overlapping show times; verify 409 Conflict
RSK-03	Privilege escalation via manipulation of role payload	Med	Critical	**High**	Auth Middleware	Disallow `role` field assignment in public registration; default strictly to `'user'`	Attempt `POST /api/auth/register` with `{ role: 'admin' }`; verify stored role
RSK-04	Plaintext password or secret key leakage in git / responses	Low	Critical	**High**	User Model / Git	Enforce Mongoose `toJSON` password deletion; add `.env` to `.gitignore`	Automated grep across codebase and commit log
RSK-05	NoSQL query injection via object injection in request body	Med	High	**Med**	Express Routers	Express-validator type checks on all string parameters	Inject `{ "$gt": "" }` payloads; verify HTTP 400 rejection
RSK-06	Orphaned booking records if seat allocation fails mid-transaction	Low	High	**Med**	Booking Service	Structure atomic single-document updates; verify update output before creating ticket	Inspect database after simulated mid-operation network failure
RSK-07	CastError crashes on malformed MongoDB ObjectIds	High	Med	**Med**	Movie/Show/Booking Routes	Parameter validation middleware verifying `isValidObjectId` before query	Query routes with `/api/movies/not-a-valid-id`; verify clean HTTP 400

---

15. Operational Setup, Environment Configuration & Repository Hygiene

15.1 Prerequisites

- **Node.js**: v18.18.0 LTS or v20.x+
- **npm**: v9.x or v10.x+
- **MongoDB**: Community Server v6.0+ running on port `27017` (or MongoDB Atlas cloud cluster)

15.2 Environment Variables Guide (`.env.example`)

Create `.env` in the backend root directory (never commit to version control):

```
# Server Network Configuration
PORT=5000
NODE_ENV=development

# Database Connection
MONGO_URI=mongodb://localhost:27017/imovies_db

# Cryptography & Token Secrets
JWT_SECRET=super_secure_jwt_secret_key_minimum_32_characters_random
JWT_EXPIRES_IN=24h
BCRYPT_SALT_ROUNDS=10

# Cross-Origin Security
CLIENT_ORIGIN=http://localhost:5173
```

15.3 Step-by-Step Local Development Startup

```
# ========================================
# 1. CLONE & NAVIGATE TO PROJECT ROOT
# ========================================
cd d:\agent2\MovieTicket

# ========================================
# 2. BACKEND SETUP & INITIALIZATION
# ========================================
cd backend
npm install
cp .env.example .env
# Edit .env with your local MongoDB URI and JWT Secret

# Start Backend Development Server (Port 5000)
npm run dev

# ========================================
# 3. FRONTEND SETUP & INITIALIZATION
# ========================================
cd ../frontend
npm install

# Start Frontend Development Server (Port 5173)
npm run dev

# Open application in browser: http://localhost:5173
```

---

16. Trackable Task Tracker Checklist

Use these trackable markdown checkboxes to monitor progress throughout the development lifecycle:

Phase 1: Environment & Project Setup

- [ ] **ENV-01**: Initialize Node.js backend package.json, Express server entry point, and directory structure.
- [ ] **ENV-02**: Implement MongoDB Mongoose connection with retry handling and event logging.
- [ ] **ENV-03**: Setup `.env.example`, security middleware (`cors`, `helmet`, `express-rate-limit`).

Phase 2: Schema Architecture

- [ ] **AUTH-01**: Define unified `User` Mongoose schema with RBAC role discrimination.
- [ ] **MOV-01**: Define `Movie` Mongoose schema with certifications, runtime, and text search indexes.
- [ ] **THTR-01**: Define `Theatre` Mongoose schema with tiered seat capacities (Lower, Middle, Balcony).

- [ ] **SHOW-01**: Define `Show` Mongoose schema with movie/theatre references, compound scheduling index, and booked seats array.
- [ ] **BOOK-01**: Define `Booking` Mongoose schema with reference UUID, seat array, and status enum.
- [ ] **FAV-01**: Define `Favorite` Mongoose schema with unique compound index `(user, movie)`.
- [ ] **REV-01**: Define `Review` Mongoose schema with unique compound index `(user, movie)` and rating validation.

Phase 3: Authentication & Security

- [ ] **AUTH-02**: Implement `bcrypt` password hashing and verification utility helpers.
- [ ] **AUTH-03**: Implement JWT token generation and verification services with expiration.
- [ ] **AUTH-04**: Build `POST /api/auth/register` customer registration endpoint.
- [ ] **AUTH-05**: Build `POST /api/auth/login` authentication endpoint.
- [ ] **AUTH-06**: Implement `fetchUser` authentication middleware.
- [ ] **AUTH-07**: Implement `authorizeRole` administrative access guard middleware.
- [ ] **AUTH-08**: Build `GET` and `PUT` `/api/auth/profile` endpoints.

Phase 4: Movies & Catalog Discovery

- [ ] **MOV-02**: Build `GET /api/movies` with search, genre filters, and pagination.
- [ ] **MOV-03**: Build `GET /api/movies/:id` movie details endpoint.
- [ ] **MOV-04**: Build `POST /api/movies` admin movie creation endpoint.
- [ ] **MOV-05**: Build `PUT` and `DELETE` `/api/movies/:id` admin endpoints.

Phase 5: Theatres & Show Scheduling

- [ ] **THTR-02**: Implement Theatre CRUD endpoints `/api/theatres`.
- [ ] **SHOW-02**: Implement `POST /api/shows` with hall schedule conflict detection.
- [ ] **SHOW-03**: Implement `GET /api/shows` queried by `movieId` and `date`.

Phase 6: Seat Booking & Concurrency

- [ ] **BOOK-02**: Build `GET /api/shows/:id/seats` live seat map and pricing API.
- [ ] **BOOK-03**: Build `POST /api/bookings` with atomic race-condition-free seat reservation.
- [ ] **BOOK-04**: Build `GET /api/bookings/my-bookings` and `GET /api/bookings/:id`.
- [ ] **BOOK-05**: Build `PUT /api/bookings/:id/cancel` booking cancellation with seat release.

Phase 7: Favorites & Community Reviews

- [ ] **FAV-02**: Build `GET`, `POST`, `DELETE` `/api/favorites` endpoints.
- [ ] **REV-02**: Build Review CRUD endpoints and automatic movie rating recalculation.

Phase 8: Frontend Client Application

- [ ] **UI-01**: Initialize React + Vite project with Axios instance and base CSS tokens.
- [ ] **UI-02**: Build `AuthContext` and token lifecycle hooks.
- [ ] **UI-03**: Build responsive Navbar, Footer, and ProtectedRoute wrappers.
- [ ] **UI-04**: Implement HomePage (`/`) and MovieCatalogPage (`/movies`).
- [ ] **UI-05**: Implement MovieDetailPage (`/movies/:id`) with showtime selector.
- [ ] **UI-06**: Implement interactive SeatBookingPage (`/book/:showId`) with tier layout.
- [ ] **UI-07**: Implement BookingConfirmationPage (`/booking/success/:id`) with printable ticket.
- [ ] **UI-08**: Implement FavoritesPage and MovieReviews component.
- [ ] **UI-09**: Implement Admin Management Portal (Dashboard, Movies, Theatres, Shows).

Phase 9: Testing & Quality Assurance

- [ ] **QA-01**: Execute automated Jest/Supertest integration test suite.
- [ ] **QA-02**: Execute automated concurrency test verifying double-booking prevention.
- [ ] **QA-03**: Conduct security audit (vulnerabilities, CORS, rate limiting, injection).

Phase 10: Deployment & Handover

- [ ] **DEP-01**: Verify production build, Docker setup, and complete final evaluation checklist.

---

17. Future Enhancements & Scalability Roadmap

1. **Payment Gateway Integration (Stripe / Razorpay)**: Transition booking creation from immediate confirmation to a two-phase commit: initiate pending booking with a 10-minute temporary seat lock, receive webhook event upon payment capture, and transition status to `CONFIRMED`.
2. **Real-Time Live Seat Availability (Socket.IO)**: Establish WebSocket connections on the seat selection page to broadcast seat selection events in real-time to other users viewing the same screening.
3. **Automated Email Notifications (Nodemailer / SendGrid)**: Asynchronous background worker queuing email delivery of booking confirmations with attached PDF tickets.
4. **Microservices Decomposition (Long-Term)**: As traffic scales beyond 10,000 requests/minute, decouple the monolithic service into specialized microservices: `auth-service`, `catalog-service`, `booking-engine`, and `notification-service` coordinated via Apache Kafka or RabbitMQ.

---

18. Glossary & Architectural Assumptions

- **Atomic Operation**: A database operation that completes entirely or has no effect at all, preventing intermediate states during concurrent access.
- **Bcrypt**: Adaptive cryptographic key derivation function based on the Blowfish cipher used for secure password hashing.
- **JWT (JSON Web Token)**: Compact URL-safe token format representing claims securely transferred between client and server.
- **RBAC (Role-Based Access Control)**: Restricting system access to authorized users based on designated organizational roles (`user` vs `admin`).
- **Referential Integrity**: Consistency between coupled database entities enforced via canonical MongoDB ObjectIds.
- **Screening Tier**: Segregated hall seating sections (Lower, Middle, Balcony) characterized by distinct pricing structures.

End of Master Project Plan Document.